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BIO302 MIDTERM MCQS

1. is estimated to have about_____super-coiled loops
- 120
- 643
- 400
- 250

2. The fist step is the initial binding of polymerase to a promoter to form what is called a_____
- Close complex
- Transription start site
- Transription bubble
- Replication fork

3. tRNA accounts for some_____of all cellular RNA
- 40%
- 30%
- 20%
- 10%

4. Polycyclic aromatic hydrocarbons (PAHs) formed by the incomplete combustion of the burning wood or____
- Carbon
- Glass
- Aluminium
- Coal

5. The_____ model predicts that the linker DNA forms a straight path between successive nucleosomes.
- Solinium
- Zigzag
- Solenoid
- Doom Bell

6. tRNA having amino acid joins with____during protein synthesis.

mRNA

rRNA

miRNA

ribosomes

7. Hershay and Chase used_____isoptop for their experiments
- N

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- P
- B
- Cd

8. In a molecule of double-stranded DNA, the amount of adenine present is a always equal to the amount of

- Cytosine
- Guanine
- Thymine
- Uracil

9. Cell free fermentation was discovered in____

- 1938
- 1865
- 1897
- 1997

10. The wave length range of UV-B light is_____nm.

- 296-320
- 100-295
- 321-400
- 350-410

11. Direct dealkylation reactions have probably been most extensively studied in____

- E. coli
- Frog
- Bat
- Human

12. The bacterium used in Hershay and Chase experiments were_____

Staphyllococus aureus

- E. coli
- Tuberculosis mycobacterium
- Sterptococcus pneumonia

13. The nucleosome of eukaryotic genome contains about_____protein molecules with DNA.

- 16
- 9
- 7
- 12

14. The wavelength of ultra violet light that damages DNA is____nm

- 100-400

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- 500-600
- 550-600
- 430-540

15. Which three additional enzyme are required when the nick is present on 3' end of the mismatch.

- MutS-alpha.Exol and RPA
- RPA, Exol and RCF
- MutL -alpha, PCNA and RCF
- MutL -alpha, PCNA and RPA

16. Presence of 5 origins of replication will divide each chromosome into_____replicons

- 5
- 2
- 0
- 6

17. RNA Polymerases function only in the presence of_____

- DNA
- Either DNA or RNA
- RNA
- None

18._____ number of distinct DNA glycosylase/lyase enzymes have been identified in human cells.

- 3
- 5
- 7
- 6

19. Which shows the correct complimentary base pairing of DNA?

- C-A, T-G
- A-G, C-T
- C-G, U-A
- T-A, G-C

20. Long patch repair pathway replace the_____nucleotides

- 2-8
- 3-18
- 2-14
- 4-20

21. Ribose and deoxyribose sugar in nucleic acids differ in_____at_____

- N, C3
- O, C2

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- H, C3
- O, C3

22. E. coli's chromosome consists of approx._____nucleotide pairs.
- 4 Million
- 5 Million
- 6 Million
- 9 Million

23. All organisms that have a mismatch repair system have MutS and_____homologs
- MutL
- MutM
- MutN
- MutS

24. Formation of cyclobutene pyrimidine dimer (CPD) which accounts for about____% of all the UV-induced damage.
- 35
- 57
- 75
- 42

25. DNA replication is semi-conservative was proved in____
- 1958
- 1955
- 1943
- 1947

26. Nucleoside is_____
- Nucleotide -Phoshoric acid
- Nucleotide + Phoshoric acid
- Nucleotide - Phoshoric acid+ N.B
- Nucleotide -Phoshoric acid+ N.B

27. The bond angles in protein primary structure resulting from rotations at C are labelled (phi) for the____
- Alpha C-O
- N -alpha C
- alphaC – C
- Alpha N – C

28. Cisplatin is a very effective chemotherapeutic agent for treating cancers of the_____
- Bladder
- Testicles
- Ovaries

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- All given

29. The fist enzyme diastase was discovered by_____
- Friedrich
- Morgan
- Miescher
- Payen

30. Key substrates required for the synthesis of DNA to proceed are____and____
- NADPH -Primer
- E. coli -Bacteriophage
- Deoxynucleoside triphosphates -Primer
- Primer -template

31. DNA polymerases introduce about one mispaired nucleotides
- 105
- 107
- 103
- 101

32. Performed the X-ray diffraction analysis of DNA fibers.
- Rosalind and Wilkins
- Crick and Avery
- Summer and Buchner
- Meselson & Franklin

33. Exol enzyme degrades DNA in a_____direction
- CH
- 5'……3'
- 3'……5'
- PO4-2

34. _____carries the information from DNA to the ribosomal sites of protein synthesis
- rRNA
- tRNA
- mRNA
- siRNA

35. Symmetry in quaternary structure of protein with two fold rotational axis is called_____
- Dihedral
- Bilateral
- Cyclic
- All given

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36. RNA create cavities on their outer surfaces specific for the different_____
- Amino acids
- Nucleotides
- ATP
- Proteins

37. _____make RNA using appropriate segments of chromosomal DNA as their templates
- DNA polymerases
- Endonucleases
- RNA polymerases
- All given

38. The fist discovered enzyme is known as_____
- Lypase
- Amylase
- Diastase
- Isomerase

39. _____adds a deoxyribonucleotide to the 3' OH end of DNA
- DNA polymerase
- Deoxyribose phosphate lyase
- Endo nuclease
- Topoisomerase

40. The movement of DNA helicase along single stand DNA strand in defined direction is known as_____
- Velocity
- Polarity
- Heterogenity
- Processivity

41. An AP (Apurinic and apyrimidinic) site can be produced by_____enzyme
- DNA glycosylase
- DNA degydrogenase
- DNA isomerase
- All given

42. Meselson and Stahl performed a classic experiment to distinguish among three possible models

Of DNA replication using isotope_____

- C
- N

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- S
- Na

43. ____is sulphur containing amino acid.
- Tryptophan
- Glutamine
- Methionine
- Histidine

44. CH3 group is attached at____of thymine nitrogenous base.
- C6
- C5
- N3
- N6

45. Psoralen can_____DNA if photo-activated
- Alkylate
- Ubiquitinate
- Phosphorylate
- Methylate

46. Which one of these amino acids are present in high perecentage in histone protein.
- Proline
- Leucine
- Lycine
- Phenyalanine

47. One of the most mutagenic base analogs is 5-bromouracil, an analog of_____
- Cytosine
- Adenine
- Thymine
- Guanine

48. The basic nucleotide excision repair pathway is_____in all organisms.
- Same
- Different
- In equilibrium
- None

49. The length of mRNA molecule is variable and depends on the length of_____
- DNA
- Protein
- rRNA
- tRNA

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50. James Watson and francis Crick discovered the double helical structure of DNA_____.
- 1926
- 1955
- 1953
- 1943

51. Which one of these amino acids are present in high perecentage in histone protein.
- Arginine
- Leucine
- Tryptophan
- Phenyalanine

52. Central dogma of molecular biology does not involve_____.
- Protein
- Carbohydrates
- DNA
- RNA

53. CPD photolyases are present in all organisms except_____
- Animals
- Bacteria
- Plants
- Humans

54. Urea was synthesized Wohler in_____.
- 1932
- 1828
- 1869
- 1833

55. The incorporation rate of rNTPs is_____-folds lower than dNTPs.
- 100
- 500
- 1000
- 1200

56. Which of the following is not a base of RNA?
- Guanine (G)
- Cytosine (C)
- Adenine (A)
- Thymine (T)

57. Which enzyme prevents the accumulation of supercoils ahead of the replication fork?
- Helicase

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- Topisomerase
- DNAse
- RNase

58. Working independently cracked the code in the early
    - Meselson and Stahl
    - Nirenberg and Gobind
    - Breedle and Tautum
    - Summer and Buncher

59.How initiation of replication takes place was explained by_____.
    - Fredrich Preston
    - Francois Jacob
    - Frank Preston
    - Joseph Connell

60. Okazaki fragment is formed during____process of cell.
    - Restriction
    - Transcription
    - Replication
    - Manipulation

61.The length RNA primer for DNA polymerase is_____nucleotides.
    - 5-10
    - 15-20
    - 20-30
    - 22-28

62.Each loop of E. coil chromosome has an average length of about____kbp
    - 30-50
    - 10-20
    - 60-65
    - 70-80

63.The replication process of DNA produces two new strands of DNA molecule, is____model of replication.
    - Semiconservative
    - Conservative
    - Dispersive
    - Seggregation

64.The innermost phosphate of a nucleotide triphosphate is known as____
    - Alpha
    - Beta
    - Gamma
    - Eta

65.Bond formation between two nucleotide is a____type of reaction.

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- SN1
- SN2
- SN3
- SN4

66.How many classes of histones?

- 8
- 3
- 4
- 5

67.DNA helicases are____proteins that assume the shape of a ring

- Octameric
- Hetameric
- Hexameric
- pentameric

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